

Object-Detection from Multi-View remote sensing Images: A case study of fruit and flower detection and counting on a central Florida strawberry farm

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ABSTRACT

Object detection in remote sensing images is one of the most critical computer vision tasks for various earth observation applications. Previous studies applied object detection models to orthomosaic images generated from the SfM (Structure-from-Motion) analysis to perform object detection and counting. However, some small objects that are occluded from the vertical view but observable in raw images from the oblique views cannot be detected in the orthomosaic image, leading to an occlusion issue that cannot be resolved with the traditional orthophoto-based approach. Taking strawberry detection as a case study, the objective of this study is to detect small objects directly from multi-view raw images. Firstly, an object-detection model (Faster R-CNN in this study) was applied to each raw image to identify strawberry fruit and flower objects. Each unique strawberry object on the ground can be detected multiple times in the raw images because images have forward- and side overlap. To find the unique objects from the step one detection results, an improved FaceNet model was proposed to combine the image and position information to calculate the feature distance between those objects, and a clustering algorithm was used to associate the cluster with each unique strawberry using the object distance output from the FaceNet model, from which the final position and number of strawberry fruits and flowers were obtained. Compared with the orthomosaic image alone, this approach using multi-view images effectively solved the occlusion problem and improved overall recognition accuracy of strawberry flowers, unripe fruits, and ripe fruits from 76.28% to 96.98%, 71.64% to 99.09%, and 69.81% to 97.17%, respectively, highlighting the potential of multi-view stereovision (MVS) in small object detection.

1. Introduction

Object detection has long been considered a common but challenging task in remote sensing and computer vision as it requires locating objects in a given image and classifying each object (Zou et al., 2019; Jiang et al., 2022). Over the past decade, various remote sensing systems have been developed for agricultural tasks, such as ground-based platforms, low-altitude UAVs (Unmanned aerial vehicles), and multirobot systems (integration of ground and aerial platforms) (Weiss et al., 2020; Ju et al., 2022). These platforms were adopted to collect high-quality and

redundant image data for agricultural robotics and high-throughput phenotyping tasks (Fountas et al., 2020; Clemente et al., 2021). Object detection is a fundamental and critical problem for these applications (Wesner et al., 2021).

In recent years, deep learning has been widely used to address the object detection problem with its powerful feature learning and extraction capabilities for remote sensing applications (Osco et al., 2021; Han et al., 2022; Wesner et al., 2021; Zhou et al., 2022). The mainstream object detectors can be divided into three categories: one-stage, two-stage, and transformer-based detectors (Zaidi et al., 2022).

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One-stage detectors simplify detection as regression problems and directly predict bounding box and class attribute based on the feature map. YOLO (You only look once) (Shafiee et al., 2017), SSD (Single Shot MultiBox Detector) (Liu et al., 2016), RetinaNet (Lin et al., 2017) and EfficientDet (Tan et al., 2020) are representative one-stage object detectors. Two-stage detectors typically generate an arbitrary number of region proposals on feature maps, which may contain objects of interest, and then perform classification and bounding box refinement on the region proposals to localize objects. Commonly used two-stage detectors include Faster R-CNN (Girshick (2015)), Cascade-RCNN (Cai and Vasconcelos (2018)) and Detectors (Qiao et al., 2021), etc.

Transformers were initially used in the natural language processing and have recently been introduced to computer vision tasks (Han et al., 2020). It relies on the self-attention mechanism which achieves global receptive field and information exchange between image samples (Liu et al., 2021). Some popular transformer-based detectors include DETR (DEtection TRansformer) (Carion et al., 2020), ViDT (Vision and Detection Transformers) (Song et al., 2021), and etc. Combined with high-resolution remote sensing images with sufficient object details, these deep learning algorithms have made considerable strides in small object detection tasks and achieved promising results even in the face of complex unstructured agricultural environments (Han et al., 2022; Reedha et al., 2022; Parico and Ahamed (2021); Mostafa et al., 2021).

Here we take the strawberry farm application as an example. Strawberry (*Fragaria × ananassa*) is one of the most popular berry fruits in the world due to its attractive appearance, delicious taste, high nutritional value, and wide production (Weng et al., 2020; Zheng et al., 2021). Since strawberry fruits are highly perishable and sensitive to the environment (Savini et al., 2005), timely and effective monitoring of fruit growth is vital for growers and marketers to determine optimal strategies for strawberry harvesting, transportation, and marketing (Abdel-Mawgoud et al., 2010; Zhou et al., 2020). Accurate forecast can help reduce waste and avoid situations where overripe fruits are not picked due to insufficient labor (Guthman 2017; Chen et al., 2019). Deep learning has been widely studied and applied in autonomous robotic harvesting and yield prediction of strawberries (Koirala et al., 2019; Vasconez et al., 2020).

Many state-of-the-art frameworks, such as Faster R-CNN, SSD, and Mask R-CNN, have been proven effective in various scenarios and complex situations of strawberry detection, such as multi-fruit adhesion, occlusion of stems and leaves, and changes in the light intensity of the environment, especially changes caused by shadows. For example, Lin and Chen (2018) compared the performance of R-CNN, Fast R-CNN, and Faster R-CNN in identifying outdoor-field grown strawberry flowers of various morphological characteristics, such as sheltered, concatenated, non-carpel, shadowed, blazed, white/pink bud-shaped, and those in blurred images. The average detection rate of the above three models were 63.4%, 76.7% and 86.1%, respectively. Lamb and Chuah (2018) implemented modifications to the SSD framework and network to achieve optimal execution speed and detection rate, which can run quickly on mobile low-power hardware for strawberry detection with an average precision of 84%. Chen et al. (2019) used Faster R-CNN to detect and count mature/immature strawberries and flowers for yield prediction from drone images taken from 2 m and 3 m heights. The mean average precision (mAP) was 83% at 2 m and 72% at 3 m height for all kinds of detected objects.

Yu et al. (2019) applied Mask R-CNN to generate mask images of ripe strawberry fruits and presented a visual localization method to get the strawberry picking points from these masks. The average detection precision rate was 95.78% and the picking point error was within 1.2 mm, which is a valuable advancement for fruit picking robots. Zhou et al. (2020) improved Faster R-CNN for greenhouse strawberry detection and plumpness evaluation by using ResNet-50 model as the backbone for feature extraction. Transfer learning was used to reduce the number of samples required for the training process, reaching a fruit extraction accuracy of about 86%. Wang et al. (2022) introduced a DSE-

YOLO (Detail- Semantics Enhancement You Only Look Once) model to detect multi-stage strawberries. The average mAP (mean average precision) and F1-score were 86.58% and 81.59%, respectively.

However, most of the studies mentioned above are based on the orthomosaic or 2D planar image, which introduces a serious problem as explained below. When using remote sensing images for fruit yield prediction, the general idea is to process a series of raw images with overlap into an orthomosaic image over the entire study area through the Structure-from-Motion (SfM) photogrammetry analysis. The deep neural networks are then applied on the orthomosaic image to detect, evaluate, and count fruits and flowers. The orthomosaic image is dominated by pixel information from one angle (usually the angle most close to vertical direction) or contains a combination of pixel information from different images taken from different directions; therefore, some strawberry objects that are occluded by leaves or blurred (due to multiple image fusion) in the orthomosaic image could be clearly visible in the oblique-angled raw images. As shown in Fig. 1, strawberries invisible from the vertical observation pose (t) could be seen from oblique angle shots ($t-1, t+1$). Compared with two-dimensional images, multi-view stereovision (MVS) may help find more strawberry fruits and flowers through the analysis of overlapping images taken from multiple perspectives. The multi-view information has been shown to substantially improve landcover mapping accuracy compared to using only the orthomosaic (Liu and Abd-Elrahman (2018a); Liu and Abd-Elrahman (2018b); Liu et al., 2019), but there are yet no studies that have attempted to use multi-view information for small object detection and counting tasks. This study proposes a new method for detecting and counting strawberry fruits and flowers directly from raw images based on deep learning and SfM technology. Initially, Faster R-CNN was applied on each raw image to detect fruits and flowers. An improved FaceNet model, and Threshold and DBSCAN clustering algorithms were then used to cluster duplicate strawberry objects obtained from raw images to get the final position and count of strawberry fruits and flowers.

2. Methodology

2.1. Experiment workflow

An outline of the experiment workflow is illustrated as Fig. 2. To evaluate the performance of our proposed approach against the conventional approach, two experiments were carried out for comparison: 1) Benchmark approach. Faster R-CNN was used to identify unripe fruits, ripe fruits, and flowers, and then make a count from the orthomosaic image; 2) Proposed approach. Faster R-CNN was first applied on each individual raw image to obtain the location information and image patches of strawberry fruits and flowers; then the duplicate strawberries detected from multi-view images were projected onto the ground using a photogrammetric model to form object (i.e., fruits or flowers) clusters; this is followed by applying clustering algorithms to find the final position and number of unique strawberry fruits and flowers.

2.2. Image acquisition

Images used in this study were collected during the 2017–2018 strawberry growing season at a strawberry farm (located in 27°45'40"N, 82°13'40"W) at the University of Florida's Gulf Coast Research and Education Center (GCREC) in Wimauma, FL, as shown in Fig. 3(a). The strawberry field included six adjacent 100 m-long raised beds, which were cultivated according to commercial standards. A Vegetation Mobile Mapping System (Fig. 3(b)) introduced by Abd-Elrahman et al. (2011, Abd-Elrahman et al. (2016)) was utilized to capture strawberry RGB images. The imaging sensor in this study was a Nikon D-300 consumer-grade digital camera. The images taken by this camera were synchronized to a Global Positioning System (GPS) receiver's time based on a custom-built electronic board. The exposure position of each image

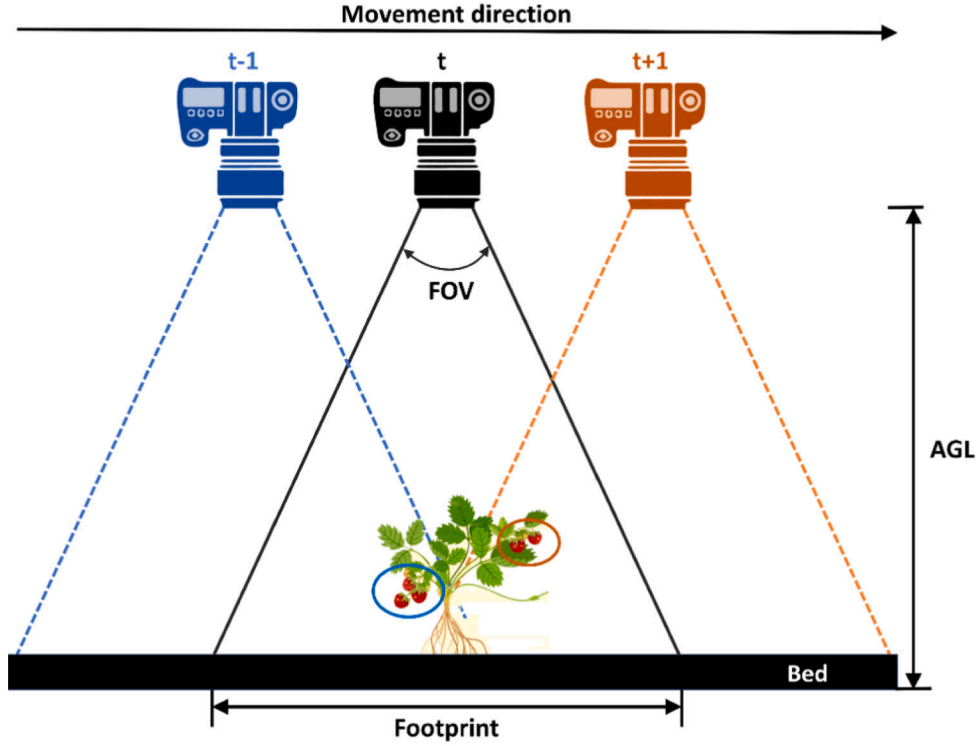


Fig. 1. Multi-view observation of strawberry fruits from ground-based images.

was determined from Global Navigation Satellite System (GNSS)-trajectory acquired by a dual-frequency Topcon HiperLite Plus GNSS receiver. This system was towed by a tractor and was placed approximately 3.5 m above the ground. The platform acquired images with more than 70% overlap and 60% sidelap. Several reliable ground control points (GCPs) with centimeter-level accuracy ($\sim 2\text{cm}$) were evenly established and distributed in the field as described by Guan et al. (2020) using a total station and GNSS receivers.

2.3. Orthomosaic and orthorectified images construction from SfM analysis

SfM, combined with photogrammetry, is a widely used technology that can be used to estimate the three-dimensional structure of an object or scene based on a series of sequential overlapping two-dimensional images (Fonstad et al., 2013). The GCPs established in the field were visible in the image and used for the SfM image processing. The process starts with the detection of tie points, image coordinate locations of features identifiable in multiple images. This is usually achieved using a proprietary feature-based image matching algorithm similar to the scale invariant feature transform (SIFT) method (Lowe (1987)). Then, a bundle-block adjustment (Triggs et al., 2000) is performed using these tie points and GCPs to compute the exterior orientation parameters (position and angular orientation) of each camera exposure. Next, a dense 3D point cloud is created over the whole scene which may be used to derive a 3D terrain surface raster or mesh. Using the estimated camera poses and parameters along with the 3D model, pixel information for a synthesized orthomosaic covering the entire area is computed, simulating orthographic projection at some nominal scale and thus removing image-displacing effects of camera tilt and terrain relief. In this study, an RGB orthomosaic image with spatial resolution of 1 mm was created and used for the strawberry detection task, shown in Fig. 2. This procedure was implemented using the Agisoft Metashape software.

For a detected strawberry object in the raw image, the relationship between the pixel coordinates of the center point position and its real

geographic coordinates is shown as Equation (1), which is derived from collinearity equations and has been used in Liu and Abd-Elrahman (2018a); Liu and Abd-Elrahman (2018b) for multi-view object extraction for mapping task. Based on Eq (1), the calculation of the real-world coordinates of a given point can theoretically be achieved by the multi-image ray intersection method with known internal and exterior elements of multiple cameras as observations (Zhang and Elaksher (2012); Yang et al., 2021). In our study, we used the Agisoft Metashape Python API to project the image coordinates of the object on the raw image in terms of row and column number to its position in the ground coordinate system in terms of (X, Y, Z) . This process comprises finding the intersection of a ray originating from the camera projection center passing through the object's image location on the raw image with the 3D model surface generated by the dense 3D point cloud. Then, the (X, Y, Z) positions obtained here are used as feature inputs for the FaceNet model as discussed in Section 2.5.

$$\begin{cases} x_p = (x_0 - f \frac{r_{11}(X - X_0) + r_{12}(Y - Y_0) + r_{13}(Z - Z_0)}{r_{13}(X - X_0) + r_{23}(Y - Y_0) + r_{33}(Z - Z_0)})/p \\ y_p = H - (y_0 - f \frac{r_{21}(X - X_0) + r_{22}(Y - Y_0) + r_{23}(Z - Z_0)}{r_{13}(X - X_0) + r_{23}(Y - Y_0) + r_{33}(Z - Z_0)})/p \end{cases} \quad (1)$$

In Eq (1), x_p, y_p are the column and row number of a given point referenced to the top left corner of the raw image; X, Y, Z represents the point coordinates in the World Coordinate System (WGS 84/UTM zone 17 N in this study); f is the focal length of camera lens in millimeters; x_0, y_0 are the principal point coordinates in millimeters; p is the pixel pitch in millimeters; H are the row dimensions of the image; X_0, Y_0, Z_0 represents the position of the camera in the World Coordinate System; r_{ij} represents the i_{th} and j_{th} element of rotation matrix. The X_0, Y_0, Z_0 coordinates, and R rotation matrix are camera exterior orientation parameters outputted from the bundle adjustment.

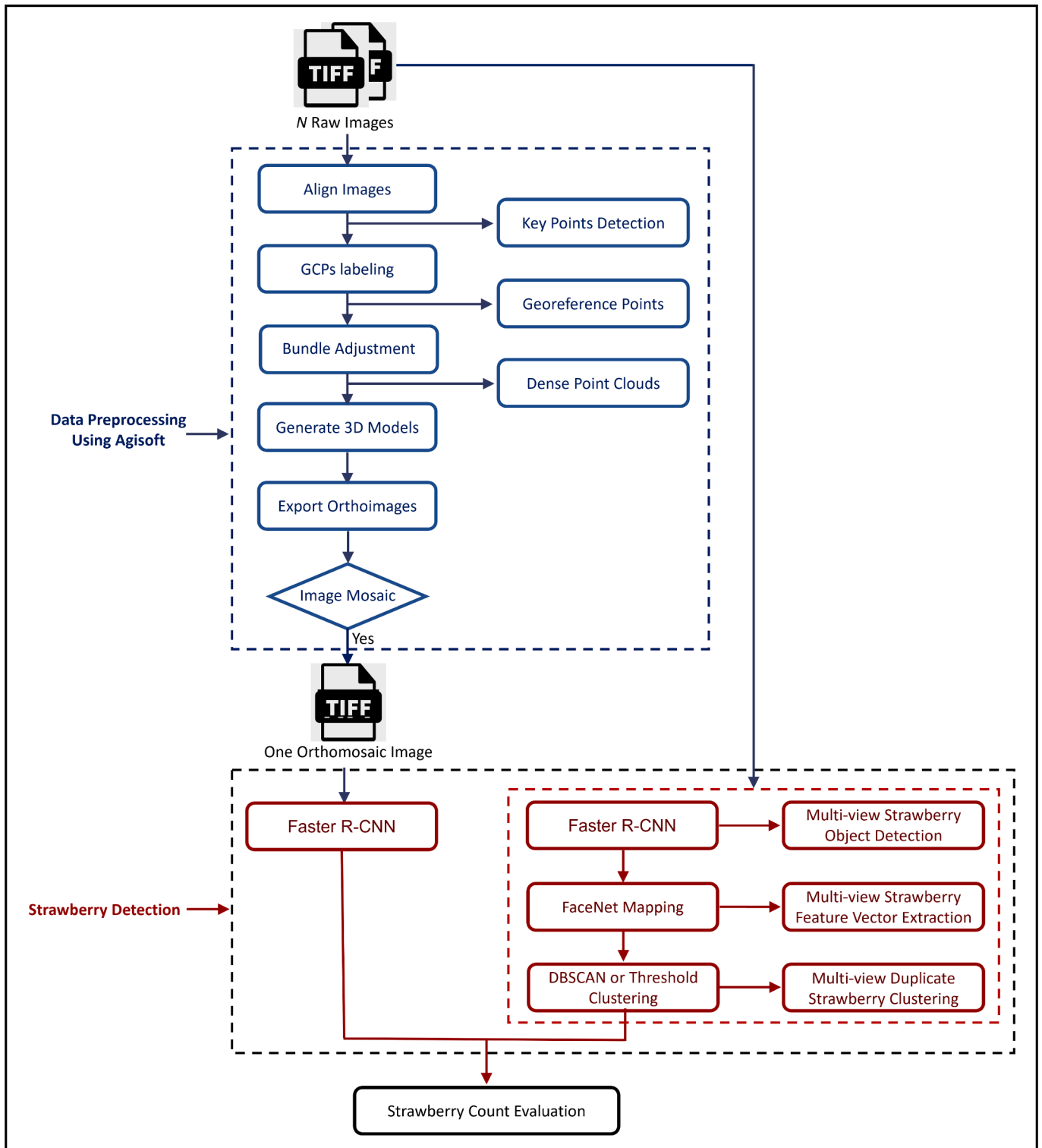


Fig. 2. Workflow of strawberry detection from multi-view high-resolution imagery.

2.4. Faster R-CNN

2.4.1. Model description

Faster R-CNN is an integrated convolutional object detection framework proposed by Ren et al. (2015). Compared with Fast R-CNN, it replaced the selective search method as the Region Proposal Network (RPN) to overcome the running time bottleneck in the task of generating proposals. The Faster R-CNN detection framework architecture first

passes a color (usually RGB) image through a pre-trained CNN using ImageNet dataset to obtain a convolutional feature map. The image feature extraction backbone used in this study is ResNet-50. Then it can be divided into two parts. The first part is to use the RPN to produce high-quality rectangular regions (bounding boxes) that could contain the object(s) of interest. The second part uses Fast R-CNN as classifier to determine if the proposals generated from RPN belong to an object class of interest. The classifier can be 2 class or N class. In this work, we

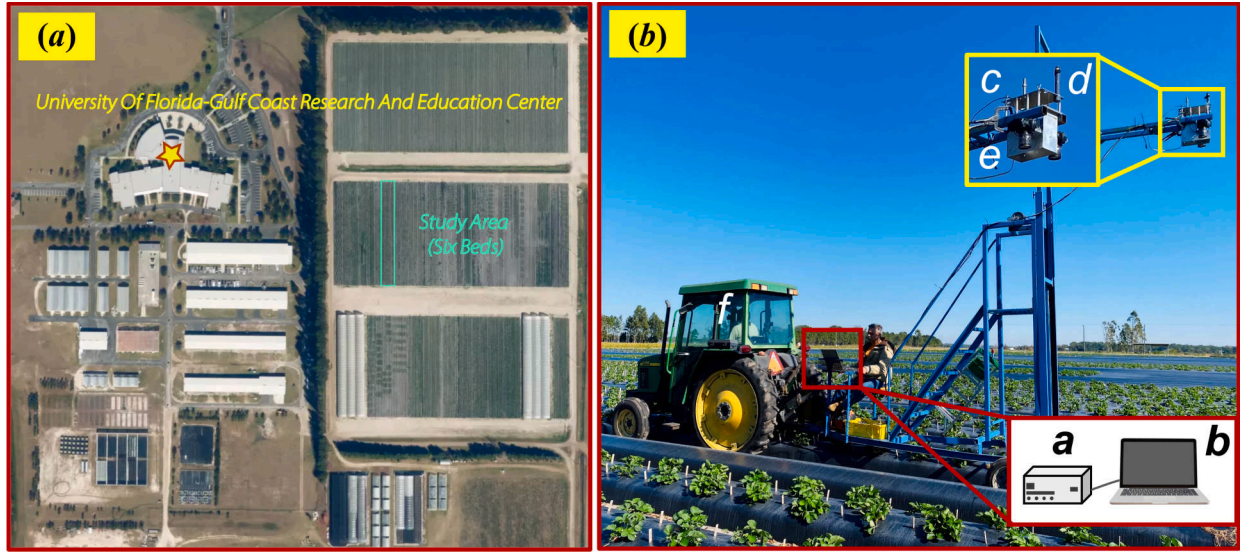


Fig. 3. The study site (a) and ground-based platform (b), which is pulled by a tractor carrying: a. camera trigger box hardware; b. computer; c. timing GPS; d. GNSS navigation system; e. Nikon D300 RGB camera.

adopted a four-class scheme composed of: flowers, unripe strawberry fruits, ripe fruits, and background.

2.4.2. Dataset and evaluation

In the training phase of the Faster R-CNN implementation, we performed fine-tuning and prepared several manually labelled (annotated) bounding box samples for strawberry flowers, ripe and unripe fruits, respectively. The numbers of training and test images are shown in Table 1.

The intersection-over-union (*IoU*) overlap with manually annotated bounding box was used to evaluate the correctness of detected objects, which was defined as follows:

$$IoU = \frac{A_O}{A_U} \quad (2)$$

where A_O indicates the overlap area between the predicted bounding box and manually annotated bounding box, and A_U is the union area of the predicted and manually annotated bounding boxes. A detected object with *IoU* higher than a threshold value was called as a true positive (TP). Otherwise, it was considered as a false positive (FP) as it did not match with any ground truth bounding box. If one manually annotated case was missed, a false negative (FN) was determined. The *IoU* threshold value was set as 0.5 in this study.

To evaluate the performance of Faster R-CNN, the precision and recall were used, which were calculated as follows:

$$\begin{cases} precision = \frac{TP}{TP + FP} \\ Recall = \frac{TP}{TP + FN} \end{cases} \quad (3)$$

2.5. FaceNet

2.5.1. FaceNet model description

FaceNet is a unified framework proposed by Schroff et al. in 2015 to

solve tasks such as face recognition, verification, and clustering. It maps each face image to a Euclidean space where the distance between feature descriptors of images reflects the face similarity. Compared with previous techniques, this model trained a deep convolution network (CNN) directly to optimize the embedding itself instead of using an intermediate bottleneck layer. In the training process, face patches are imported to the CNN network and each patch includes three images: anchor, positive, and negative. Considering the small training samples, the CNN feature extraction backbone used in this study is ResNet-34 as shown in Fig. 5, which was also adopted in the work of Schroff et al. 2015. The anchor and positive images belong to the same identity while the negative image is a different identity (see Fig. 4). Thus, a triplet loss (L) was employed and minimized, and it aims to train the embedding model that makes the squared distance between all faces, regardless of imaging conditions, of the same identity is small, while the distance between images of different identities is large. The triplet loss can be calculated as following formulas:

$$L = \sum_{i=0}^N \left[\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha \right] \quad (4)$$

here, $f(x)$ is an embedding model that maps an image x into a feature space R^d . The length of the output feature vector is set as 128. x_i^a , x_i^p , and x_i^n represent the anchor, positive and negative image of the i^{th} patch, respectively; N is the number of image patch; α represents the margin between positive and negative pairs.

2.5.2. Improved FaceNet

To utilize multi-view observations to identify potential strawberries occluded by leaves, this study applied Faster R-CNN to all raw images for strawberry detection. The orthomosaic image was used as a benchmark for comparison. Due to the overlap between the raw images, many duplicate strawberries corresponding to one cluster (or one strawberry identity) were detected from different views, as shown in Fig. 6. It is necessary to cluster these strawberry objects to obtain the final number, location, and distribution of strawberries. Before the clustering process, we improved the FaceNet model to directly learn a mapping from images and geographical position of strawberry objects to a compact Euclidean space. As shown in Fig. 5, two models were presented to provide each strawberry object a feature vectorial representation, known as embedding: 1) Model-1: according to the bounding box information provided by Faster R-CNN, a cropped image of a detected strawberry object was

Table 1
Faster R-CNN training samples preparation.

	Training	Test
Number of Images	2058	357
Size	536 × 712	

Table 2

Faster R-CNN performance on the orthomosaic image.

Category	Manually Count	Faster R-CNN count (TP + FP)	TP	TP Rate	FP	FP Rate	FN	FN Rate	P	R
Flower	346	328	313	90.5%	15	4.3%	33	9.5%	95.4%	90.5%
Unripe Fruit	417	394	374	89.7%	20	4.8%	43	10.3%	94.9%	89.7%
Ripe Fruit	78	74	71	91.0%	3	3.8%	7	8.97%	95.9%	91.0%
All	841	796	758	90.1%	38	4.5%	83	9.87%	95.2%	90.1%

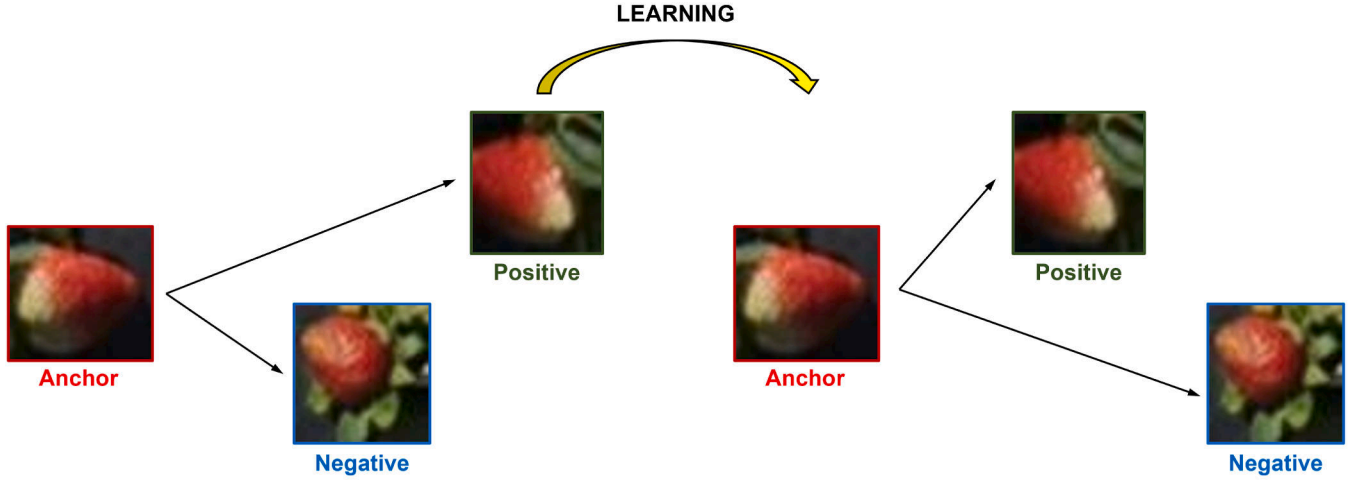


Fig. 4. The anchor and positive image pair belong to the same identity while the anchor and negative image are from different identities. The triplet loss in FaceNet mapping aims to minimize the distance between the anchor and positive and maximize the distance between the anchor and negative image. The length of the black arrow indicates the distance.

imported to the FaceNet model. Similar to the face verification discussed in section 2.5.1, images of detected objects were mapped to feature vectors and the distance between image-derived vectors were used to assess if they belong to the same strawberry identity based on a given threshold; 2) Model-2: in addition to the images, the position information (X, Y, Z) of image center were also used in the output feature vector of the deep architecture.

2.5.3. Labeled strawberry dataset

To train the FaceNet model, we prepared a dataset of manually labeled images for strawberry flowers, unripe and ripe fruits, respectively. In this study, Faster R-CNN was applied on 456 raw images and detected 4363 objects, which were manually corrected and interpreted, and corresponded to 1086 unique strawberry identities. The image samples of these detected objects were cropped and saved according to the bounding box. Each strawberry object was marked with the following attributes: $X, Y, Z, Sample_ID, Strawberry_ID, True_Class, IF_Visible$, and $Photo_ID$. Here, X, Y and Z are the geographic coordinates of the centroid point of the bounding box. $Sample_ID$ is the image sample name. $Strawberry_ID$ represents which strawberry identity it belongs to. $True_Class$ is the sample class (flower, unripe fruit, ripe fruit). $IF_Visible$ indicates if this strawberry object is visible in the orthomosaic image, where 1 represents True, and 0 represents False. $Photo_ID$ represents the raw image name from which it was detected. Due to the difficulty of ground-truth sample preparation, we selected 995 samples of 250 strawberry identities in a small area for FaceNet model training and set aside 500 samples for testing. Then, the FaceNet model and the clustering process was implemented on all 4363 samples to calculate the number of strawberry objects. Examples of strawberry sample data are shown in Fig. 7.

2.5.4. FaceNet model performance evaluation

Given a pair of two strawberry objects (i, j), the calculated distance D

(x_i, x_j) using the FaceNet model was compared with a given squared L_2 distance threshold d . Then we can determine a binary classification of *same* and *different*. All strawberry pairs (i, j) of the same identity are denoted with ρ_{same} while the pairs of different identities were denoted with ρ_{dif} .

The set of *true positive* (TP) represents these strawberry pairs (i, j) correctly classified as *same* according to the threshold d , which is defined as follows:

$$TP(d) = \{(i, j) \in \rho_{same}, \text{ with } D(x_i, x_j) \leq d\} \quad (5)$$

The set of *false positive* (FP) represents those pairs which were falsely classified as *same* at the threshold d . $FP(d)$ is defined as:

$$FP(d) = \{(i, j) \in \rho_{dif}, \text{ with } D(x_i, x_j) \leq d\} \quad (6)$$

Similarly, we define the set of *true negative* (TN) and *false negative* (FN) as:

$$\begin{cases} TN(d) = \{(i, j) \in \rho_{dif}, \text{ with } D(x_i, x_j) > d\} \\ FN(d) = \{(i, j) \in \rho_{same}, \text{ with } D(x_i, x_j) > d\} \end{cases} \quad (7)$$

where TN and FN are the pairs (i, j) that were correctly and falsely classified as *different* at the threshold d , respectively.

To assess the FaceNet model performance, the validate rate $VAL(d)$, false accept rate $FAR(d)$ and accuracy (d) based on the threshold given by the model are defined as:

$$\begin{cases} VAL(d) = \frac{TP(d)}{\rho_{same}}, FAR(d) = \frac{FP(d)}{\rho_{dif}} \\ Accuracy_{FaceNet} = \frac{TP(d) + TN(d)}{TP(d) + TN(d) + FP(d) + FN(d)} \end{cases} \quad (8)$$

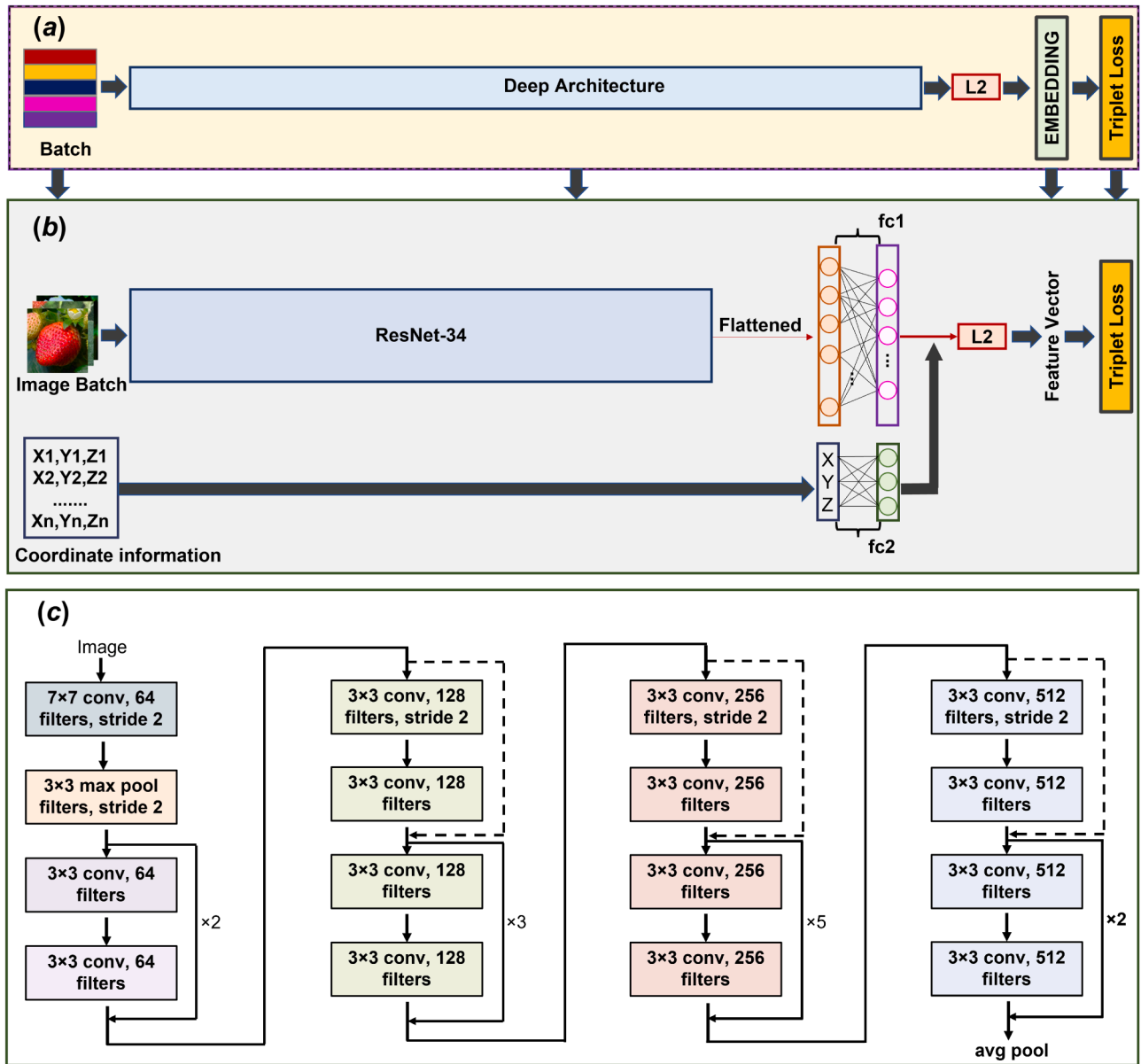


Fig. 5. Model structure. (a) FaceNet: this network imports an image batch to a deep CNN followed by L_2 normalization to get an embedding feature vector, which was used in the triplet loss training; (b) Improved FaceNet: it combines the image and positional information to obtain an embedding feature vector for each strawberry object. The backbone network used in this study was ResNet-34; (c) ResNet-34 architecture.

2.6. Strawberry clustering

2.6.1. Threshold clustering

The threshold clustering approach was used in the work of Schroff et al. (2015) to achieve face clustering. It calculates the Euclidean distance between a pair of faces based on the FaceNet embedding feature vectors. When the distance between feature vectors is smaller than a given threshold, the strawberry instances are assigned to the same cluster. Thus, the embedding of a new instance is compared with already clustered strawberry instances. If the distance between a new strawberry fruit or flower instance with its nearest neighbor already in a cluster set is less than a specified threshold, it will be added to the existing cluster. Otherwise, it is considered a new cluster. The performance of this clustering algorithm heavily relies on the selection of the threshold. A low threshold will result in many false negatives: instances with a distance higher than the threshold but belong to the same strawberry

identity. High threshold will cause many false positives: instances having a distance lower than the threshold but coming from different strawberry identities. In our experiment, we used the optimal threshold with highest validation accuracy in the improved FaceNet training process as the distance criterion. Each detected strawberry instance was evaluated and clustered based on the improved FaceNet embedding.

2.6.2. Density based spatial clustering of applications with noise (DBSCAN)

DBSCAN is a density-based data clustering algorithm proposed by Ester et al. (1996). DBSCAN is a non-parametric parametric method that doesn't need to specify the exact number of clusters in advance (Khan et al., 2014). It can cluster data of arbitrary shapes and is also robust in the presence of noise. The DBSCAN algorithm performs clustering based on two input parameters: 1) *eps* specifies the maximum neighborhood distance threshold (radius) for a given point, 2) *minpts* denotes the minimum number of points lying in the radius *eps* of one point to form a



Fig. 6. A set of strawberry clusters shown in the orthomosaic image at two zoom scales (a-b). One unique strawberry cluster (or identity) contains several centroid points of bounding boxes detected from multi-view raw images using Faster R-CNN model. The yellow points represent strawberry flowers. The cyan points and dark pink points represent strawberry unripe and ripe fruits, respectively. Several cases show up in (b) where objects are obscured by leaves on orthomosaic but are detected on multi-view raw images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

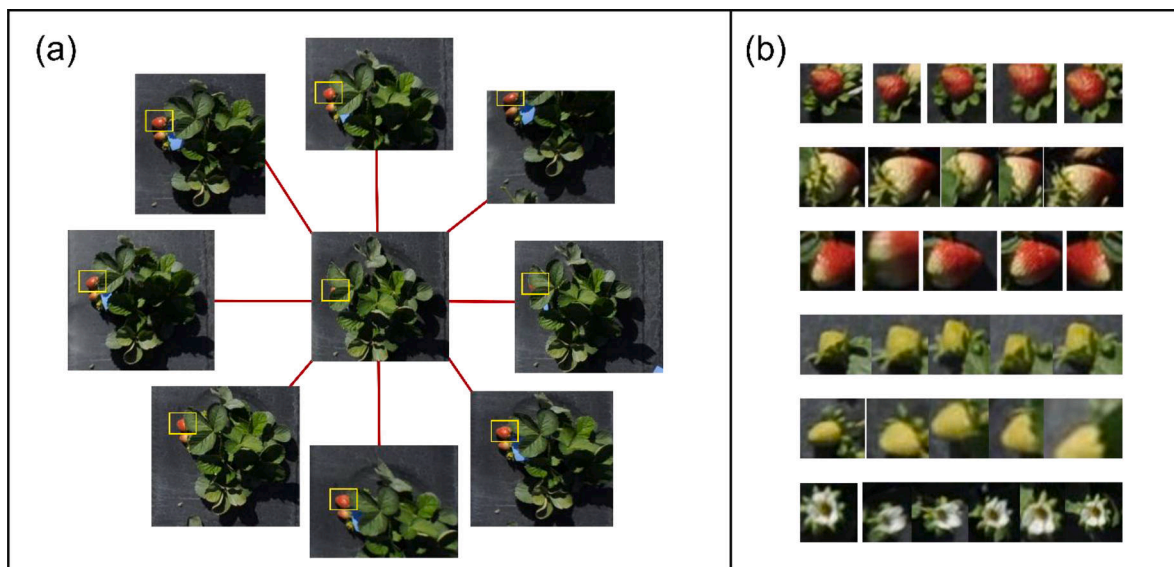


Fig. 7. Strawberry samples detected by Faster R-CNN. (a) Multi-view observations of one strawberry identity. The central image came from the orthomosaic image, and the other images were from raw images; (b) Samples obtained from raw images. Each row represents one strawberry identity.

cluster. According to the two parameters, several concepts were defined:

- *eps-neighborhood*: for any point $p \in D$, the *eps-neighborhood* (N_{eps}) denotes the total number of points whose distance from p is less than or equal to eps . D represents the database of points.

$$N_{eps} = \{q \in D, \text{withdist}(p, q) \leq eps\} \quad (9)$$

- *Core points*: If the *eps-neighborhood* of a point p contains more than $minpts$ points, it is considered as core point. The core point is defined as:

$$N_{eps}(p) > minpts \quad (10)$$

- *Border points*: A point that is lying at the border of a core point's *eps* neighborhood is called as border point.
- *Noise points*: If a point does not belong to any cluster, it is treated as a noise point.
- A point q is said to be directly density reachable from a point p if $q \in N_{eps}$ and p is a core point.
- If there is a sequence of points $q_1 \dots q_n$, $q_1 = p$, $q_n = q$, such that q_{i+1} is direct density-reachable from q_i , the point q is considered density reachable from p .
- A point p is density-connected to another point q if there is one core point o such that p and q are both density reachable from o .

DBSCAN checks the *eps-neighborhood* of each point in the point database. If N_{eps} of a point p contains at least $minpts$ points, p is a core point and all the points of $N_{eps}(p)$ are created as a new cluster X . Then, for each point $q \in N_{eps}$, the *eps-neighborhood* will be checked. If q is also a core point, all the points of $N_{eps}(q)$ which are not already assigned will be added to the cluster X . This procedure continues until no new point can be added to the cluster X .

This study applied the threshold clustering and DBSCAN algorithms on the improved FaceNet embedding of duplicated strawberry objects to perform clustering. Beside the distance principle, another criterion is that if two objects have the same *Photo_ID*, they will not be clustered to the same group. This is based on the hypothesis that all the detected objects from the same image are unique.

2.6.3. Clustering evaluation

To evaluate the performance of two clustering algorithms, the pairwise F-measure defined by Bijl (2018) is used in our experiments. Assuming two sets of labels: L_{True} and $L_{Cluster}$. The set $L_{True} = \{t_1, t_2, \dots, t_n\}$ contains the actual identity labels for each detected strawberry object. Set $L_{Cluster} = \{c_1, c_2, \dots, c_n\}$ is the output label of the clustering algorithm for each strawberry object.

The set of true positives ($TP_{Cluster}$) consists of strawberry object pairs (i, j) that are correctly clustered to the same cluster, which is defined as:

$$TP_{Cluster} = \{(i, j), \text{where } c_i = c_j \text{ and } t_i = t_j\} \quad (11)$$

The set of false positives ($FP_{Cluster}$) consists of strawberry object pairs (i, j) that are incorrectly clustered to the same cluster, which is defined as:

$$FP_{Cluster} = \{(i, j), \text{where } c_i = c_j \text{ and } t_i \neq t_j\} \quad (12)$$

The set of true negative ($TN_{Cluster}$) consists of strawberry object pairs (i, j) that are correctly clustered to different clusters, which is defined as:

$$TN_{Cluster} = \{(i, j), \text{where } c_i \neq c_j \text{ and } t_i \neq t_j\} \quad (13)$$

The set of false negatives ($FN_{Cluster}$) consists of strawberry object pairs (i, j) that are incorrectly clustered to different clusters, which is defined as:

$$FN_{Cluster} = \{(i, j), \text{where } c_i \neq c_j \text{ and } t_i = t_j\} \quad (14)$$

The pairwise precision ($Precision_{Cluster}$) is defined as the proportion of pairs that are correctly clustered to the same cluster ($TP_{Cluster}$) over all pairs that were clustered to the same cluster by the cluster algorithm ($TP_{Cluster} + FP_{Cluster}$). Therefore, the $Precision_{Cluster}$ is defined as:

$$Precision_{Cluster} = \frac{TP_{Cluster}}{TP_{Cluster} + FP_{Cluster}} \quad (15)$$

Similarly, the pairwise recall ($Recall_{Cluster}$) the proportion of pairs that are correctly clustered to the same cluster ($TP_{Cluster}$) over all pairs that are actually of the same cluster ($TP_{Cluster} + FN_{Cluster}$). Therefore, the pairwise recall is given by:

$$Recall_{Cluster} = \frac{TP_{Cluster}}{TP_{Cluster} + FN_{Cluster}} \quad (16)$$

Finally, the F-measure can be defined as:

$$F = 2 \frac{Precision_{Cluster} \cdot Recall_{Cluster}}{Precision_{Cluster} + Recall_{Cluster}} \quad (17)$$

2.7. Overall performance evaluation

With the number of strawberries manually counted from raw images as reference, the overall strawberry detection accuracy of different models including: 1) Faster R-CNN applied in the orthomosaic image, 2) Faster R-CNN, FaceNet and Threshold Clustering, and 3) Faster R-CNN, FaceNet and DBSCAN Clustering, can be defined as follows:

$$Overall \text{ Accuracy} = 1 - \frac{|N_R - N_p|}{N_R} \quad (18)$$

In Eq (18), N_R is the reference number of unique strawberries, which manually counted in this study by visual interpretation of multi-view raw images; N_p is the predicted number of strawberries detected from the above three models.

3. Results

All our experiments were implemented under the deep learning framework of Pytorch 1.11, with an Nvidia TITANX (Pascal generation) for GPU acceleration. The strawberry detection experiments from the orthomosaic image and multi-view raw images were carried out separately, and the results are as follows:

3.1. Orthomosaic image analysis

A total number of 236 raw images (2848×4288 pixels) were processed using the Agisoft software to generate one orthomosaic image over the whole field. Faster R-CNN was then implemented on the orthomosaic image to detect flowers, unripe fruits, and ripe fruits. Fig. 8 shows strawberry flower and fruit detection examples from the orthomosaic image. The detection performance of Faster R-CNN is summarized in Table 1. The precision and recall rate are 95.4% and 90.5% for flowers, 94.9% and 89.7% for unripe fruits, and 95.9% and 91.0% for ripe fruits, respectively. The average precision and recall rate for all three types of strawberry objects were 95.2% and 90.1%, respectively. The detection accuracy of unripe fruits was lower than that of ripe fruits and flowers, possibly due to the similar color of the unripe fruit to the leaves.

3.2. Multi-view image analysis

3.2.1. FaceNet mapping

The classification (same or different) was performed on the prepared strawberry dataset using the FaceNet model as discussed in Section 2.5.3. Fig. 9 illustrates the training process of the FaceNet model. After one hundred epochs, the training and validation classification accuracy can reach about 85.8% and 86.1% using Model-1, 99.62% and 99.1%



Fig. 8. Faster R-CNN detection examples form the orthomosaic image.

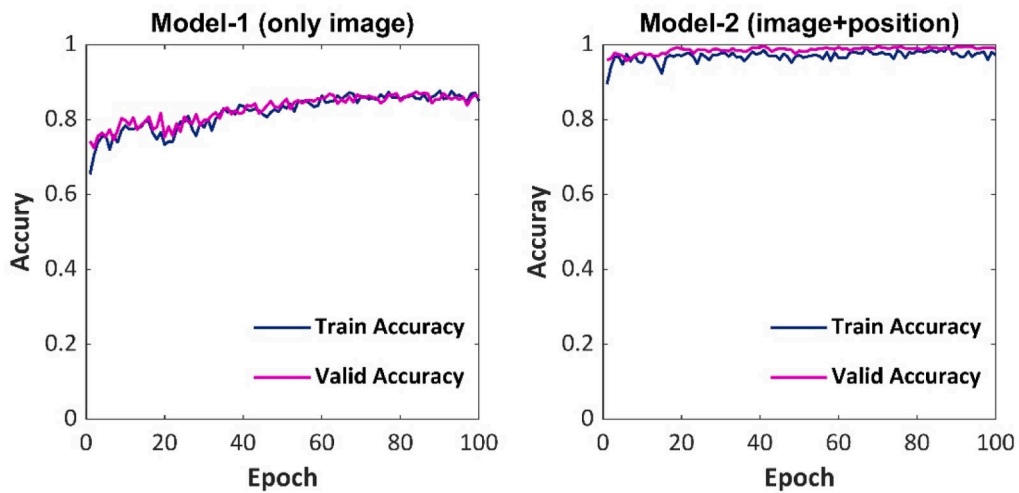


Fig. 9. Accuracy curve for two models during the training process.

using Model-2, respectively. Table 3 shows the FaceNet model performance on the testing set of strawberry flowers, unripe and ripe fruits. Compared with Model-1, the Model-2 increased the VAL of flowers from 84.63% to 98.36%, that of unripe fruits from 83.75% to 95.29%, and that of ripe fruits from 86.8% to 99.68%, respectively. Correspondingly, the Model-2 decreased the FAR of flowers from 10.31% to 1.11%, that of unripe fruits from 9.53% to 1.08%, and that of ripe fruits from 8.42% to

0.25%, respectively. The addition of geographic information substantially improved the model performance in determining whether a pair of detected strawberry objects belong to the same identity or not. The results of FaceNet on ripe fruit are slightly better than the other two types of objects, probably because ripe strawberries have richer color and shape information.

Table 3

Model performance of the testing set of strawberry flowers, unripe and ripe fruits.

Category	FaceNet Model	VAL	FAR
Flowers	Model-1	84.63%	10.31%
	Model-2	98.36%	1.11%
Unripe Fruits	Model-1	83.75%	9.53%
	Model-2	95.29%	1.08%
Ripe Fruits	Model-1	86.8%	8.42%
	Model-2	99.68%	0.25%

3.2.2. Clustering

The optimal FaceNet model during the training process was applied on all detected duplicate strawberry samples to obtain the embedding feature vectors. Then, we used two methods to cluster strawberries based on these features: threshold clustering and DBSCAN algorithm. Table 4 summarizes the clustering performance of the two algorithms. For threshold clustering, the F-measure values for flowers, unripe fruits, and ripe fruits are 0.853, 0.870, and 0.891, respectively. The corresponding F-measure values of DBSCAN algorithm are 0.860, 0.869 and 0.904, respectively. The difference between predicted and known clusters is not obvious, and the precision and recall metrics fluctuate between 0.791 and 0.973. Fig. 10 shows the variation curve of F-measure values and the number of strawberry clusters with different distance threshold settings (d in Section 2.5.4). With increasing the distance, strawberries are clustered together. Increasing the distance makes less and large clusters. As shown in Fig. 10 (a-e), there was a narrow range of distance threshold values which achieved the highest F-measure values and optimal number of strawberry clusters. Both algorithms were able to detect the number of clusters close to the ground truth with comparable accuracy (Table 4), but the DBSCAN algorithm appeared to be slightly more effective and robust than threshold clustering.

Table 5 shows the comparison results of the number of strawberry objects counted by human visual interpretation and deep learning from the orthomosaic image and raw images. The occlusion rate is defined as the ratio of the number of strawberry objects invisible in the orthomosaic image to the number of strawberries counted from raw images. In this experiment, the occlusion rate was about 19.53%, 24.18%, and 26.42% for strawberry flowers, unripe fruits, and ripe fruits, respectively. The average occlusion rate was about 22.56%. Some occlusion examples are shown as Fig. 11. Based on Table 5, the FaceNet clustering algorithms based on the raw images effectively solved the occlusion problem and improved the overall recognition accuracy of strawberry flowers, unripe fruits, and ripe fruits from 76.28% to 96.98%, 71.64% to 99.09%, and 69.81% to 97.17% respectively, compared with the results obtained from the orthomosaic image using Faster R-CNN only.

4. Discussion

Counting and locating flowers and fruits in the strawberry field from remote sensing images is traditionally conducted by applying deep learning networks (such as Faster R-CNN and YOLO series) on an orthomosaic image of the whole study area generated by SfM analysis for object detection. This method has been proven effective in many previous studies (Lin and Chen (2018); Lamb and Chuah (2018); Chen

et al., 2019). However, some fruits that are obscured by leaves but observable in raw images could be lost in the orthomosaic image due to photogrammetric processes such as pixel merging and blending. As shown in Table 2, although Faster R-CNN exhibits high detection rates (~90.1%) in the orthomosaic image, this accuracy is still only around 73.3% based on the actual number of strawberries on the ground. Therefore, the occlusion issue is the main reason affecting the recognition accuracy rather than the deep learning object detection model itself. This study aims to develop a novel method to make full use of the information in the multi-view raw images acquired by a ground-based platform to solve the occlusion problem.

4.1. General interpretation

Our proposed method mainly consists of three steps: 1) strawberry object detection from raw images using Faster R-CNN; 2) clustering feature extraction using FaceNet; 3) duplicate strawberry detection removal using clustering algorithm. We tested the off-the-shelf algorithms for step 1, 2 and 3, and developed alternative customized algorithms to carry out step 2.

In step 1, we adopted a commonly used deep learning model named Faster R-CNN for object detection. It showed good performance in both orthomosaic and multi-view image (see Table 3 and 6). However, with the overall occlusion rate up to 22.56%, the traditional approach based on orthomosaic image won't obtain comparable accuracy compared to the proposed approach even if it can detect all the visible objects on the orthomosaic image. Since the proposed approach is based on the object detection results by Faster R-CNN, poor detection of objects will certainly damage the performance of the proposed approach. Table 6 indicates that occlusion is the key factor preventing the traditional approach from catching up with the proposed approach and Faster R-CNN is not a bottleneck due to its good performance for object detection task.

In step 2, the off-the-shelf algorithm FaceNet was used to learn feature embeddings that will be used in step 3 for clustering task. Alternatively, we also developed a new feature learning neural network based on FaceNet by adding geographic location information of strawberry objects (see Fig. 5). Our experiment shows by adding geographic location information, the feature learning algorithm achieved a substantially high improvement to separate objects (e.g., 1.11% vs 10.31% FAR for flowers, 9.53% vs 1.08% FAR for unripe fruits in Table 3). The embedding generated from step 2 will be directly used in step 3 to discriminate unique objects among the duplicate objects which were generated from step 1 from multi-view raw images; therefore, a robust embedding extractor is important to achieve good clustering performance in step 3.

In step 3, we tested two clustering algorithms including threshold and DBSCAN to find the unique objects. In Fig. 10 (a-e), we analyzed the impact of the threshold parameter on the performance of the thresholding clustering algorithm. Our results indicate that there is a narrow range of distance threshold values that can achieve the highest F-measure scores and optimal number of strawberry clusters. The F-measure scores for the threshold clustering algorithm were 0.853, 0.870, and 0.891 for flowers, unripe fruits, and ripe fruits, respectively. For the DBSCAN algorithm, the corresponding F-measure scores were 0.860,

Table 4

Performance of two clustering algorithms on the strawberry flowers, unripe and ripe fruits.

Target	Clustering method	Number of predicted clusters	Number of known clusters	Precision _{Cluster}	Recall _{Cluster}	F-measure
Flowers	Threshold clustering	417	430	0.872	0.837	0.853
	DBSCAN	414		0.791	0.943	0.860
Unripe fruits	Threshold clustering	585	550	0.839	0.905	0.870
	DBSCAN	555		0.860	0.879	0.869
Ripe fruits	Threshold clustering	109	106	0.821	0.973	0.891
	DBSCAN	109		0.850	0.965	0.904

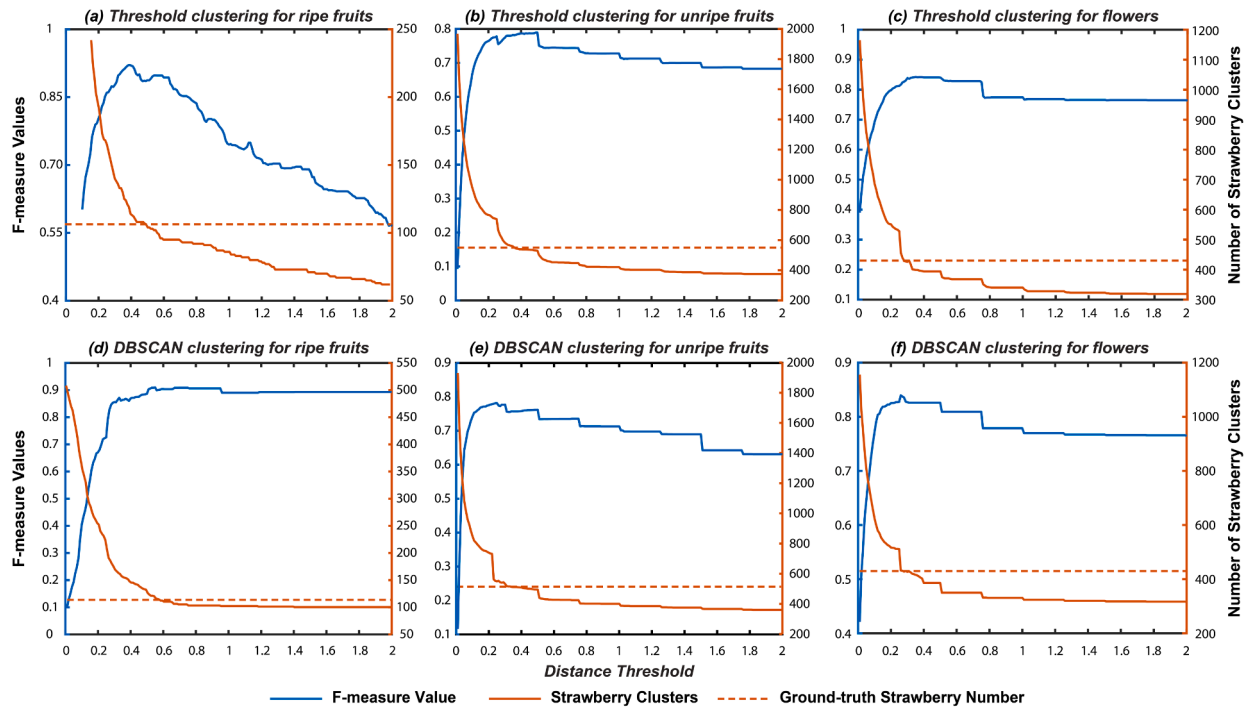


Fig. 10. The variation curve of F-measure values and strawberry clusters with the distance threshold.

Table 5

Comparison of the number of strawberries detected from the orthomosaic image and raw images using human visual interpretation and deep learning technology, respectively.

Category	Manually counted		Occlusion rate	Counted by deep learning		
	From orthomosaic image	From raw images		From orthomosaic image	From raw images DBSCAN	Threshold clustering
Flowers	346	430	19.53%	328	414	417
Unripe Fruits	417	550	24.18%	394	555	585
Ripe Fruits	78	106	26.42%	74	109	109
All	841	1086	22.56%	796	1078	1111

0.869, and 0.904, respectively. Both algorithms were able to detect the number of clusters close to the ground truth with comparable accuracy (Table 4), but the DBSCAN algorithm appeared to be slightly more effective and robust than threshold clustering. It's worth noting that the accuracy of the clustering outputs could be influenced by several factors, including the choice of clustering algorithm, the quality of feature embedding, and the accuracy of object detection. Nevertheless, our study suggests that robust algorithms are available to achieve good performance for each task, and combining these algorithms can lead to good performance for the final results.

4.2. Limitations and future work

This study has yet to test the performance of our method during a growing stage when almost all the strawberries become mature, and the strawberry farm is crowded with strawberries growing together. The proposed method involves object detection, feature learning and clustering steps, indicating that it may be challenging to achieve reliable recognition results for such a scenario. To enhance the method's performance, other advanced deep learning object detection models such as YOLO and SSD can be explored, and a unified model integrating FaceNet feature mapping and clustering procedures could be developed in the future. In addition, the improvement of image data quality may contribute to the reliable model performance of our approach. The study hypothesizes that there could be an optimal oblique viewing angle for maximizing the clear view of occluded strawberries without losing

information from the vertical angle. Future research will focus on designing a more optimized imaging platform that can capture images from multiple views and generate detailed 3D reconstructions of strawberry objects. The overlap rate setting will also impact the number of multi-angle observations for object detection, with a larger overlap generating more information but increasing algorithm complexity and model robustness challenges. Investigating the effect of side/front overlap is another potential direction for future research. Despite its limitations, this study is the first attempt to directly extract information about small objects like strawberry fruits and flowers from raw multi-view images.

5. Conclusion

Taking strawberry detection as a case study, we proposed a new method to detect small objects directly from the raw images containing multi-view information with comprehensive use of deep learning and SfM technologies. This differs from traditional methods that rely on object detection on orthomosaic images and are not well-suited to deal with occluded objects. The result shows the raw images helped us find 19.53%, 24.18%, and 26.42% more flowers, unripe fruits, and ripe fruits respectively than the orthomosaic image. Correspondingly, the use of FaceNet clustering algorithms in raw images effectively solves the occlusion problem and increased the overall recognition accuracy from 76.28% to 96.98%, 71.64% to 99.09%, and 69.81% to 97.17% for those three types of strawberry objects, in contrast to using only Faster R-CNN

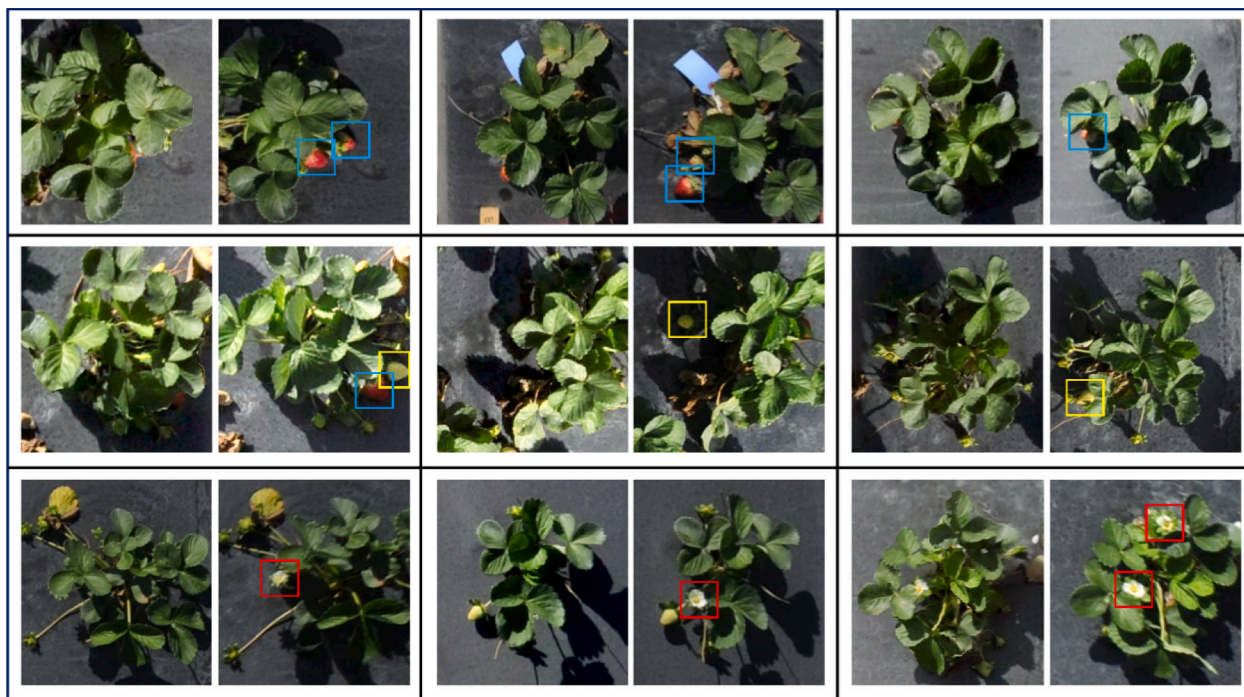


Fig. 11. Examples of strawberries obscured by leaves in the orthomosaic image and but visible in raw images: flowers with red bounding boxes, unripe fruits with yellow bounding boxes, and ripe fruits with blue bounding boxes. Nine samples are displayed. For each sample, the unlabeled image on the left is cropped from the orthomosaic image, and the right image is from raw images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 6

The overall accuracy of deep learning models applied in the orthomosaic image and raw images.

	From orthomosaic image	From raw images	
		DBSCAN	Threshold clustering
Flowers	76.28%	96.28%	96.98%
Unripe Fruits	71.64%	99.09%	93.64%
Ripe Fruits	69.81%	97.17%	97.17%
All	73.30%	99.26%	97.70%

on the orthomosaic image. This study demonstrates that multi-view object detection has the potential to outperform previous approaches in detecting occluded objects from remote sensing images.

CRediT authorship contribution statement

Caiwang Zheng: Methodology, Software, Writing – original draft, Formal analysis, Visualization, Writing – review & editing. **Tao Liu:** Conceptualization, Methodology, Software, Writing – review & editing, Supervision. **Amr Abd-Elrahman:** Conceptualization, Supervision, Project administration. **Vance M. Whitaker:** Supervision, Writing – review & editing. **Benjamin Wilkinson:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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